Terafence
Reliable Transformations!

Complete Galvanic Network Isolation

for enhanced security and protection



TFG-1URP



Technology Overview

Engineered to safeguard IoT and IIoT devices, the Terafence TFG-1URP employs Air-Gap technology to segment and isolate industrial end-devices from malicious or unauthorized malware attacks, ensuring complete protection against threats.

At the core of the solution is Terafence's proprietary FPGA-based hardware architecture, developed and manufactured by Terafence which establishes a fully controlled data path between two network segments. While allowing normal protocol data to flow unidirectionally, the return path is entirely eliminated—effectively creating an Air-Gapped environment.

Acting as a full protocol breaker / proxy server, Terafence terminates TCP/IP sessions at both ends while transferring only raw data between the two unidirectional gateway interfaces. Since raw data is never stored within the unit, there is no need for encryption or additional data protection measures. As a result, Terafence eliminates the reliance on cryptographic security mechanisms and instead enforces security by preventing network access altogether.

Beyond protecting IoT and IIoT end-devices from cyber threats, the TFG-1URP also safeguards other network assets by preventing compromised devices from spreading malware, executing cyberattacks, or causing operational disruptions.

Why choose TFG-1URP?

Key Features

- Proprietary Terafence FPGA-based hardware architecture for hardware-enforced security.
- Two dedicated CPUs (Atom E3940) for protocol support, processing, and termination.
- Secure HTTPS-based web GUI for configuration and management, accessible only from the secure side.
- Hardened operating system on the dedicated CPUs.
- Core security hardware operates independently without an operating system, MAC address, or IP address.

Solution Highlights

- Delivers physical network isolation through Terafence's proprietary FPGA-based hardware architecture.
- Supports protocol processing and termination through two dedicated proxy engines.
- Provides secure and intuitive configuration through a restricted HTTPS-based management interface.
- Designed for high availability, long-term operation, and a high MTBF (~20 years).
- Certified to IEC 62443 and MIL-STD-810F standards.

Supported Protocols

TFG-1URP supports multiple, simultaneous industrial and IT protocols, including:
SFTP/FTP/SMB – Secure file or folder-tree transfers (10GB max)
SMTP – Email relay
SYSLOG Forwarding – Log event transmission
RTSP/ONVIF – Live streaming for CCTV (10 CCTV max)
HTTP/S – File and data uploads to web servers or cloud platforms
Local SYSLOG Forwarding – Internal log event management
Modbus – Communication between multiple PLCs and HMIs
IEC-104 – Protocol for power utilities
MQTT – Data brokering and publishing
And various other protocols as per customer requirements.

Design:	Fanless architecture with no moving mechanical parts for enhanced reliability
Chassis:	IP30 protection, SECC, Black Metal
RAM / Local Storage:	8GB-DDR4/64GB-eMMC
Large-File Handling:	<10GB
Latency:	Protocol / data size dependent, core latency <2 nanoseconds

Technical Specifications

Product Type:	Unidirectional secure gateway / full protocol breaker with FPGA data diode – 19" rack version – fixed internal configuration
Data Throughput:	Up to 1 Gbps with auto-negotiation
Power Supply:	100–240 VAC, 50/60 Hz, with dual redundant internal hot-swappable AC power supplies
Network Interfaces:	2 × Ethernet (RJ-45) ports
Form Factor:	1U rack-mountable appliance
Mounting Options:	19-inch 1U rack mount
Dimensions:	(W × H × D): 440 × 44 × 260 mm (1U)
Operating Temperature:	32°F to 113°F
Storage Temperature:	-40 °F to 185 °F
Relative Humidity:	5 to 95%, 55°C (non-condensing)
Environment:	Indoor use only
Compliance:	IEC 62443, MIL-STD-810F, FCC, CE, and EMI/EMC

All images are for illustrative purposes only. Product configurations, specifications, and features may be customized to meet customer-specific requirements and may vary accordingly.

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